

Task 4

Github: https://github.com/Sirska-Mariia/homework1/tree/micro_mq

Завдання:

- Запускається *config-server*,
- Запустити *facade-service*, *counter-service* та три екземпляри *logging-service* (кожен з екземплярів запускається у власному Docker-контейнері), і далі реєструється в *config-server*,
- Відповідно має запуснутись також кластер *Hazelcast* з трьох серверів

```
PS C:\анз\homework1\banking_system> docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAME
c7370f9a2e81	my-counter-service	"uvicorn main:app --..."	7 minutes ago	Up 3 minutes	0.0.0.0:8084->8000/tcp	counter-app
62561c1e22c3	my-logging-service	"uvicorn main:app --..."	7 minutes ago	Up 3 minutes	0.0.0.0:8083->8000/tcp	logging-node-3
d799ba833ceb	my-logging-service	"uvicorn main:app --..."	7 minutes ago	Up 3 minutes	0.0.0.0:8081->8000/tcp	logging-node-1
15df8250a111	my-facade-service	"uvicorn main:app --..."	7 minutes ago	Up 3 minutes	0.0.0.0:8080->8000/tcp	facade-app
b6c9a756ea74	my-logging-service	"uvicorn main:app --..."	7 minutes ago	Up 3 minutes	0.0.0.0:8082->8000/tcp	logging-node-2
8235481d3791	postgres:15	"docker-entrypoint.s..."	7 minutes ago	Up 3 minutes	0.0.0.0:5432->5432/tcp	banking_system_db
35eabd8527f2	my-config-server	"uvicorn main:app --..."	7 minutes ago	Up 3 minutes	0.0.0.0:8000->8000/tcp	config-app
82750567e31c	hazelcast/hazelcast:5.3.0	"hz start"	7 minutes ago	Up 3 minutes	5701/tcp	banking_system_hazelcast-2
dc586adb11d6	hazelcast/hazelcast:5.3.0	"hz start"	7 minutes ago	Up 3 minutes	5701/tcp	banking_system_hazelcast-1
3a439fd1f0e7	hazelcast/hazelcast:5.3.0	"hz start"	7 minutes ago	Up 3 minutes	5701/tcp	banking_system_hazelcast-3

```
PS C:\анз\homework1\banking_system>
```

- Через *HTTP POST* записати 10 транзакцій *msg1-msg10* через *facade-service*

```
PS C:\анз\homework1\banking_system> 1..10 | ForEach-Object { curl.exe -X POST http://localhost:8080/transaction -H "Content-Type: application/json" -d "{ \"user_id\": \"Mariia\", \"amount\": $ }" }
```

```
{ "status": "Transaction accepted (queued)", "transaction_id": "6be4aae6-e67b-4ae5-8f1e-dc0217fba95f" } { "status": "Transaction accepted (queued)", "transaction_id": "bb561b86-c116-44b1-9997-8fa2967b6cf2" } { "status": "Transaction accepted (queued)", "transaction_id": "69005677-a3d2-4543-af22-c6219f7aad8e" } { "status": "Transaction accepted (queued)", "transaction_id": "c562e560-f4e6-4e0c-b134-d02b78daeaab" } { "status": "Transaction accepted (queued)", "transaction_id": "87202b6d-0fc2-4780-88b4-ed2ee960eb6d" } { "status": "Transaction accepted (queued)", "transaction_id": "2740c8c9-08c0-4d28-9871-6f7b4af77bc5" } { "status": "Transaction accepted (queued)", "transaction_id": "ebc6b076-aab2-4574-82ae-fc1f02c78b6e" } { "status": "Transaction accepted (queued)", "transaction_id": "af28eca7-fc15-4dd0-8910-2c8332f262b4" } { "status": "Transaction accepted (queued)", "transaction_id": "caaf8b8a-7f39-4964-84b3-c8373b6a38b2" } { "status": "Transaction accepted (queued)", "transaction_id": "262122e2-521c-4c00-9b06-88cd84cbf792" }
```

```
PS C:\анз\homework1\banking_system>
```

- Показати що повідомлення отримують різні екземпляри *logging-service* (це має бути видно у логах сервісу)

```
logging-node-2 | INFO:      172.19.0.7:43672 - "POST /log HTTP/1.1" 200 OK
logging-node-3 | [logging-3] Logged: bb561b86-c116-44b1-9997-8fa2967b6cf2 for user Mariia
logging-node-2 | [logging-2] Logged: 69005677-a3d2-4543-af22-c6219f7aad8e for user Mariia
logging-node-3 | INFO:      172.19.0.7:45268 - "POST /log HTTP/1.1" 200 OK
logging-node-2 | INFO:      172.19.0.7:43680 - "POST /log HTTP/1.1" 200 OK
config-app     | INFO:      172.19.0.7:56152 - "GET /nodes/logging-service HTTP/1.1" 200 OK
facade-app     | INFO:      172.19.0.1:56298 - "POST /transaction HTTP/1.1" 200 OK
logging-node-3 | [logging-3] Logged: c562e560-f4e6-4e0c-b134-d02b78daeaab for user Mariia
logging-node-3 | INFO:      172.19.0.7:45268 - "POST /log HTTP/1.1" 200 OK
config-app     | INFO:      172.19.0.7:56152 - "GET /nodes/logging-service HTTP/1.1" 200 OK
facade-app     | INFO:      172.19.0.1:56314 - "POST /transaction HTTP/1.1" 200 OK
logging-node-2 | [logging-2] Logged: 87202b6d-0fc2-4780-88b4-ed2ee960eb6d for user Mariia
logging-node-2 | INFO:      172.19.0.7:43672 - "POST /log HTTP/1.1" 200 OK
config-app     | INFO:      172.19.0.7:56152 - "GET /nodes/logging-service HTTP/1.1" 200 OK
facade-app     | INFO:      172.19.0.1:56316 - "POST /transaction HTTP/1.1" 200 OK
config-app     | INFO:      172.19.0.7:56152 - "GET /nodes/logging-service HTTP/1.1" 200 OK
facade-app     | INFO:      172.19.0.1:56318 - "POST /transaction HTTP/1.1" 200 OK
logging-node-3 | [logging-3] Logged: ebc6b076-aab2-4574-82ae-fc1f02c78b6e for user Mariia
logging-node-3 | INFO:      172.19.0.7:45268 - "POST /log HTTP/1.1" 200 OK
config-app     | INFO:      172.19.0.7:56152 - "GET /nodes/logging-service HTTP/1.1" 200 OK
logging-node-1 | [logging-1] Successfully registered at http://logging-1:8000 in Config Server
facade-app     | INFO:      172.19.0.1:56330 - "POST /transaction HTTP/1.1" 200 OK
logging-node-1 | [logging-1] Logged: 2740c8c9-08c0-4d28-9871-6f7b4af77bc5 for user Mariia
logging-node-1 | INFO:      172.19.0.7:50928 - "POST /log HTTP/1.1" 200 OK
logging-node-1 | [logging-1] Logged: af28eca7-fc15-4dd0-8910-2c8332f262b4 for user Mariia
logging-node-1 | INFO:      172.19.0.7:50932 - "POST /log HTTP/1.1" 200 OK
config-app     | INFO:      172.19.0.7:56152 - "GET /nodes/logging-service HTTP/1.1" 200 OK
facade-app     | INFO:      172.19.0.1:56346 - "POST /transaction HTTP/1.1" 200 OK
logging-node-2 | [logging-2] Logged: caaf8b8a-7f39-4964-84b3-c8373b6a38b2 for user Mariia
logging-node-2 | INFO:      172.19.0.7:43672 - "POST /log HTTP/1.1" 200 OK
config-app     | INFO:      172.19.0.7:56152 - "GET /nodes/logging-service HTTP/1.1" 200 OK
facade-app     | INFO:      172.19.0.1:56356 - "POST /transaction HTTP/1.1" 200 OK
logging-node-1 | [logging-1] Logged: 262122e2-521c-4c00-9b06-88cd84cbf792 for user Mariia
logging-node-1 | INFO:      172.19.0.7:50928 - "POST /log HTTP/1.1" 200 OK
config-app     | INFO:      172.19.0.7:53122 - "GET /nodes/logging-service HTTP/1.1" 200 OK
config-app     | INFO:      172.19.0.7:53122 - "GET /nodes/counter-service HTTP/1.1" 200 OK
logging-node-3 | INFO:      172.19.0.7:46642 - "GET /logs/Mariia HTTP/1.1" 200 OK
counter-app    | MQ Consumer Error: Heartbeat timed out
counter-app    | INFO:      172.19.0.7:45852 - "GET /balance/Mariia HTTP/1.1" 200 OK
facade-app     | INFO:      172.19.0.1:40932 - "GET /user/Mariia HTTP/1.1" 200 OK
```

```

counter app
PS C:\an3\homework1\banking_system> docker logs -f logging-node-1
INFO: Started server process [1]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO: Shutting down
INFO: Waiting for application shutdown.
INFO: Application shutdown complete.
INFO: Finished server process [1]
[logging-1] Successfully registered at http://logging-1:8000 in Config Server
INFO: Started server process [1]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO: Shutting down
INFO: Waiting for application shutdown.
INFO: Application shutdown complete.
INFO: Finished server process [1]
[logging-1] Successfully registered at http://logging-1:8000 in Config Server
INFO: Started server process [1]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
[logging-1] Successfully registered at http://logging-1:8000 in Config Server
[logging-1] Logged: 2740c8c9-08c0-4d28-9871-6f7b4af77bc5 for user Mariia
INFO: 172.19.0.7:50928 - "POST /log HTTP/1.1" 200 OK
[logging-1] Logged: af28eca7-fc15-4dd0-8910-2c8332f262b4 for user Mariia
INFO: 172.19.0.7:50932 - "POST /log HTTP/1.1" 200 OK
[logging-1] Logged: 262122e2-521c-4c00-9b06-88cd84cbf792 for user Mariia
INFO: 172.19.0.7:50928 - "POST /log HTTP/1.1" 200 OK

```

```

PS C:\an3\homework1\banking_system> docker logs -f logging-node-2
INFO: Started server process [1]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO: Shutting down
INFO: Waiting for application shutdown.
INFO: Application shutdown complete.
INFO: Finished server process [1]
[logging-2] Successfully registered at http://logging-2:8000 in Config Server
INFO: Started server process [1]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO: Shutting down
INFO: Waiting for application shutdown.
INFO: Application shutdown complete.
INFO: Finished server process [1]
[logging-2] Successfully registered at http://logging-2:8000 in Config Server
INFO: Started server process [1]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
[logging-2] Successfully registered at http://logging-2:8000 in Config Server
[logging-2] Logged: 6be4aae6-e67b-4ae5-8f1e-dc0217fba95f for user Mariia
INFO: 172.19.0.7:43672 - "POST /log HTTP/1.1" 200 OK
[logging-2] Logged: 69005677-a3d2-4543-af22-c6219f7aad8e for user Mariia
INFO: 172.19.0.7:43680 - "POST /log HTTP/1.1" 200 OK
[logging-2] Logged: 87202b6d-0fc2-4780-88b4-ed2ee960eb6d for user Mariia
INFO: 172.19.0.7:43672 - "POST /log HTTP/1.1" 200 OK
[logging-2] Logged: caaf8b8a-7f39-4964-84b3-c8373b6a38b2 for user Mariia
INFO: 172.19.0.7:43672 - "POST /log HTTP/1.1" 200 OK

```

```
PS C:\анз\homework1\banking_system_> docker logs -f logging-node-3
INFO: Started server process [1]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO: Shutting down
INFO: Waiting for application shutdown.
INFO: Application shutdown complete.
INFO: Finished server process [1]
[logging-3] Successfully registered at http://logging-3:8000 in Config Server
INFO: Started server process [1]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
INFO: Shutting down
INFO: Waiting for application shutdown.
INFO: Application shutdown complete.
INFO: Finished server process [1]
[logging-3] Successfully registered at http://logging-3:8000 in Config Server
INFO: Started server process [1]
INFO: Waiting for application startup.
INFO: Application startup complete.
INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
[logging-3] Successfully registered at http://logging-3:8000 in Config Server
INFO: 172.19.0.7:52664 - "GET /logs/Mariia HTTP/1.1" 200 OK
[logging-3] Logged: bb561b86-c116-44b1-9997-8fa2967b6cf2 for user Mariia
INFO: 172.19.0.7:45268 - "POST /log HTTP/1.1" 200 OK
[logging-3] Logged: c562e560-f4e6-4e0c-b134-d02b78daeaab for user Mariia
INFO: 172.19.0.7:45268 - "POST /log HTTP/1.1" 200 OK
[logging-3] Logged: ebc6b076-aab2-4574-82ae-fc1f02c78b6e for user Mariia
INFO: 172.19.0.7:45268 - "POST /log HTTP/1.1" 200 OK
INFO: 172.19.0.7:46642 - "GET /logs/Mariia HTTP/1.1" 200 OK
INFO: 172.19.0.7:33802 - "GET /logs/Mariia HTTP/1.1" 200 OK
□
```

- Через HTTP GET з facade-service прочитати транзакції, перевірити що значення на рахунках є коректним

← → ↻ ⓘ localhost:8080/user/Mariia

Pretty-print ☒

```
{
  "balance": 55,
  "transactions": [
    {
      "transaction_id": "87202b6d-0fc2-4780-88b4-ed2ee960eb6d",
      "user_id": "Mariia",
      "amount": 5
    },
    {
      "transaction_id": "c562e560-f4e6-4e0c-b134-d02b78daeaab",
      "user_id": "Mariia",
      "amount": 4
    },
    {
      "transaction_id": "caaf8b8a-7f39-4964-84b3-c8373b6a38b2",
      "user_id": "Mariia",
      "amount": 9
    },
    {
      "transaction_id": "6be4aae6-e67b-4ae5-8f1e-dc0217fba95f",
      "user_id": "Mariia",
      "amount": 1
    },
    {
      "transaction_id": "bb561b86-c116-44b1-9997-8fa2967b6cf2",
      "user_id": "Mariia",
      "amount": 2
    },
    {
      "transaction_id": "69005677-a3d2-4543-af22-c6219f7aad8e",
      "user_id": "Mariia",
      "amount": 3
    },
    {
      "transaction_id": "af28eca7-fc15-4dd0-8910-2c8332f262b4",
      "user_id": "Mariia",
      "amount": 8
    },
    {
      "transaction_id": "262122e2-521c-4c00-9b06-88cd84cbf792",
      "user_id": "Mariia",
      "amount": 10
    },
  ],
}
```

- Перевірка відмовостійкості:

- Вимкнути чи зробити недоступним (*docker pause*) *counter-service* та перевірити, що запис транзакцій йде без помилок (бо транзакції до *counter-service* накопичуються у *message queue*)

```
PS C:\апз\homework1\banking_system_> docker pause counter-app
counter-app
PS C:\апз\homework1\banking_system_> 
```

Curl

```
curl -X 'POST' \
  'http://localhost:8080/transaction' \
  -H 'accept: application/json' \
  -H 'Content-Type: application/json' \
  -d '{
    "user_id": "Mariia",
    "amount": 20
  }'
```

Request URL

```
http://localhost:8080/transaction
```

Server response

Code

Details

200

Response body

```
{
  "status": "Transaction accepted (queued)",
  "transaction_id": "7f8775e6-5574-4ae0-92fa-36b45fc56540"
}
```

Response headers

```
content-length: 98
content-type: application/json
date: Mon, 20 Apr 2026 22:35:50 GMT
server: uvicorn
```

Responses

Code

Description

200

Successful Response

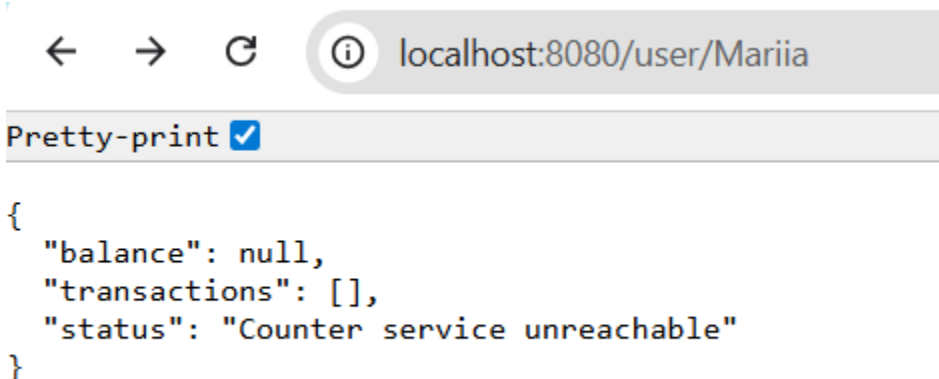
```

config-app | INFO: 172.19.0.7:46010 - "GET /nodes/counter-service HTTP/1.1" 200 OK
logging-node-2 | INFO: 172.19.0.7:53310 - "GET /logs/Mariia HTTP/1.1" 200 OK
facade-app | INFO: 172.19.0.1:38006 - "GET /user/Mariia HTTP/1.1" 200 OK
config-app | Registered: None at None
config-app | INFO: 172.19.0.1:47708 - "POST /register HTTP/1.1" 200 OK
hazelcast-1 | 2026-04-20 22:27:39,117 [ INFO] [hz.jovial_blackburn.cached.thread-4] [c.h.i.s.t.TcpServerConnection]: [172.19.0.5]:5701 [dev] [5.3.0] Connection[id=16, /172.19.0.5:5701->/172.19.0.11:47192, qualifier=null, endpoint=[172.19.0.11]:47192, remoteUid=7c932241-8245-435e-971a-8d4eed14b6f, alive=false, connectionType=PYH, planeIndex=-1] closed. Reason: Client heartbeat is timed out, closing connection to Connection[id=16, /172.19.0.5:5701->/172.19.0.11:47192, qualifier=null, endpoint=[172.19.0.11]:47192, remoteUid=7c932241-8245-435e-971a-8d4eed14b6f, alive=true, connectionType=PYH, planeIndex=-1]. Now: 2026-04-20 22:27:39.114. LastTimePacketReceived: 2026-04-20 22:22:37.621
hazelcast-1 | 2026-04-20 22:27:39,122 [ INFO] [hz.jovial_blackburn.event-1] [c.h.c.i.ClientEndpointManager]: [172.19.0.5]:5701 [dev] [5.3.0] Destroying ClientEndpoint{connection=Connection[id=16, /172.19.0.5:5701->/172.19.0.11:47192, qualifier=null, endpoint=[172.19.0.11]:47192, remoteUid=7c932241-8245-435e-971a-8d4eed14b6f, alive=false, connectionType=PYH, planeIndex=-1], clientUid=7c932241-8245-435e-971a-8d4eed14b6f, clientName=hz.client_0, authenticated=true, clientVersion=5.6.0, creationTime=1776723227070, latest clientAttributes=null, labels=[]}
hazelcast-3 | 2026-04-20 22:27:39,138 [ INFO] [hz.youthful_khayyam.cached.thread-5] [c.h.i.s.t.TcpServerConnection]: [172.19.0.6]:5701 [dev] [5.3.0] Connection[id=14, /172.19.0.6:5701->/172.19.0.11:60928, qualifier=null, endpoint=[172.19.0.11]:60928, remoteUid=7c932241-8245-435e-971a-8d4eed14b6f, alive=false, connectionType=PYH, planeIndex=-1] closed. Reason: Client heartbeat is timed out, closing connection to Connection[id=14, /172.19.0.6:5701->/172.19.0.11:60928, qualifier=null, endpoint=[172.19.0.11]:60928, remoteUid=7c932241-8245-435e-971a-8d4eed14b6f, alive=true, connectionType=PYH, planeIndex=-1]. Now: 2026-04-20 22:27:39.135. LastTimePacketReceived: 2026-04-20 22:22:37.621
hazelcast-3 | 2026-04-20 22:27:39,143 [ INFO] [hz.youthful_khayyam.event-2] [c.h.c.i.ClientEndpointManager]: [172.19.0.6]:5701 [dev] [5.3.0] Destroying ClientEndpoint{connection=Connection[id=14, /172.19.0.6:5701->/172.19.0.11:60928, qualifier=null, endpoint=[172.19.0.11]:60928, remoteUid=7c932241-8245-435e-971a-8d4eed14b6f, alive=false, connectionType=PYH, planeIndex=-1], clientUid=7c932241-8245-435e-971a-8d4eed14b6f, clientName=hz.client_0, authenticated=true, clientVersion=5.6.0, creationTime=1776723227061, latest clientAttributes=null, labels=[]}
hazelcast-2 | 2026-04-20 22:27:39,401 [ INFO] [hz.competent_booth.cached.thread-8] [c.h.i.s.t.TcpServerConnection]: [172.19.0.2]:5701 [dev] [5.3.0] Connection[id=14, /172.19.0.2:5701->/172.19.0.11:58494, qualifier=null, endpoint=[172.19.0.11]:58494, remoteUid=7c932241-8245-435e-971a-8d4eed14b6f, alive=false, connectionType=PYH, planeIndex=-1] closed. Reason: Client heartbeat is timed out, closing connection to Connection[id=14, /172.19.0.2:5701->/172.19.0.11:58494, qualifier=null, endpoint=[172.19.0.11]:58494, remoteUid=7c932241-8245-435e-971a-8d4eed14b6f, alive=true, connectionType=PYH, planeIndex=-1]. Now: 2026-04-20 22:27:39.388. LastTimePacketReceived: 2026-04-20 22:22:37.621
hazelcast-2 | 2026-04-20 22:27:39,407 [ INFO] [hz.competent_booth.event-3] [c.h.c.i.ClientEndpointManager]: [172.19.0.2]:5701 [dev] [5.3.0] Destroying ClientEndpoint{connection=Connection[id=14, /172.19.0.2:5701->/172.19.0.11:58494, qualifier=null, endpoint=[172.19.0.11]:58494, remoteUid=7c932241-8245-435e-971a-8d4eed14b6f, alive=false, connectionType=PYH, planeIndex=-1], clientUid=7c932241-8245-435e-971a-8d4eed14b6f, clientName=hz.client_0, authenticated=true, clientVersion=5.6.0, creationTime=1776723227077, latest clientAttributes=null, labels=[]}

```

Незважаючи на те, що counter-service недоступний, facade-service успішно приймає запити (і повертає статус 200). Це відбувається тому, що запити просто складаються у чергу повідомлень у Hazelcast

- При цьому, відповідно GET-запити не мають спрацьовувати, мають або повертати null у якості значень на рахунку



Оскільки сервіс лічильника та база даних заморожені, GET-запит не може отримати актуальний баланс і повертає null.

- Знову запустити counter-service та переконатись, що за деякий час він зчитує з черги накопичені до нього транзакції, застосує їх та почне повертати коректні значення рахунків

```
PS C:\апз\homework1\banking_system_> docker unpause counter-app
counter-app
PS C:\апз\homework1\banking_system_> docker restart counter-app
counter-app
```

Після виконання команди `docker unpause` сервіс лічильника відновив роботу, але оскільки пауза тривала довго, кластер Hazelcast розірвав з'єднання через відсутність heartbeat-сигналів (помилка `TargetDisconnectedError`). Для відновлення підключення до черги було виконано `docker restart counter-app`. Як видно з логів, сервіс успішно перереєструвався у Config Server, підключився до черги та вичитав транзакції, які були накопичені під час збою.

```
counter-app | INFO: Application startup complete.
counter-app | INFO: Uvicorn running on http://0.0.0.0:8000 (Press CTRL+C to quit)
config-app | Registered: counter-service at http://counter-service:8000
config-app | INFO: 172.19.0.11:34592 - "POST /register HTTP/1.1" 200 OK
hazelcast-2 | 2026-04-20 22:43:59.325 [ INFO ] [hz.competent.booth.cached.thread-1] [c.h.c
```

← → ↻ ⓘ localhost:8080/user/Mariia

Pretty-print ☒

```
{
  "balance": 75,
  "transactions": [
    {
      "transaction_id": "7f8775e6-5574-4ae0-92fa-36b45fc56540",
      "user_id": "Mariia",
      "amount": 20
    },
    {
      "transaction_id": "87202b6d-0fc2-4780-88b4-ed2ee960eb6d",
      "user_id": "Mariia",
      "amount": 5
    },
    {
      "transaction_id": "c562e560-f4e6-4e0c-b134-d02b78daeaab",
      "user_id": "Mariia",
      "amount": 4
    },
    {
      "transaction_id": "caaf8b8a-7f39-4964-84b3-c8373b6a38b2",
      "user_id": "Mariia"
    }
  ]
}
```